

**궁내교
수위 예측 AI
성능 개선 결과 3**
(시계열)

※ 강우 이벤트 전부 학습으로 사용

MIET 학습 데이터 기본 설정

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 12 (각 LSTM 3개 처리)

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 1420개 中 1412개 데이터 파일 학습데이터로 사용 (결측 데이터 5개)

검증 : 학습 데이터에서 제외한 3개 데이터 사용 ++ 학습에서 검증에 사용된 3개 데이터 사용

- 학습 : ~ 2022 68번 (988개)
- 검증 : ~ 2024 96번 (282개)
- 테스트 : ~ 2025 109번 (142개)

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 12 (각 LSTM 3개 처리)

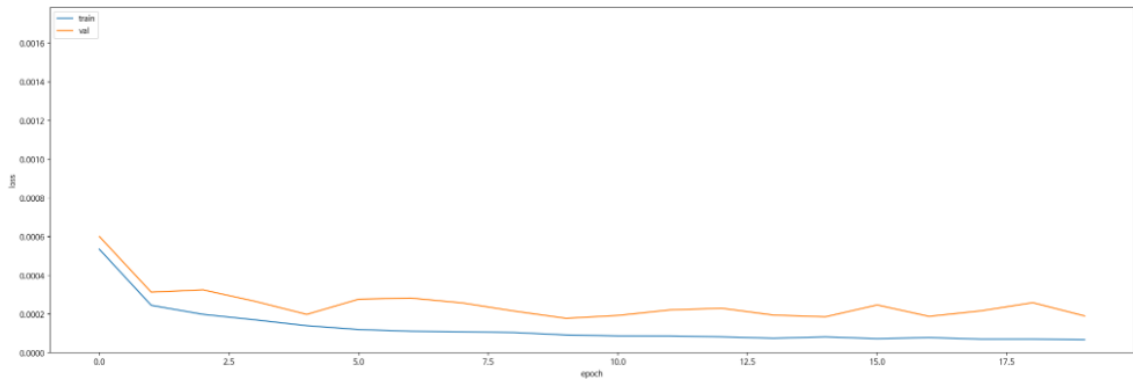
궁내교

1. num_of_layer = 18

num_of_layer =18
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

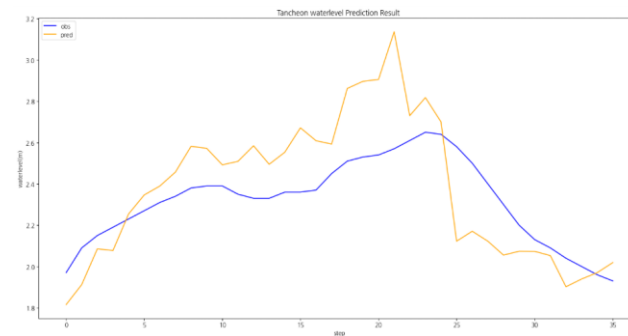
Train (23111 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0254	0.0242	0.0372	0.968928	0.970272	22.160994

Valid (6282 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0385	0.0421	0.0542	0.942801	0.951583	4.037687

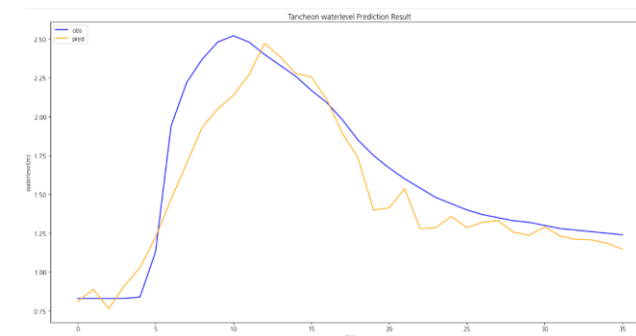
Test (5472 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0453	0.0528	0.0625	0.890020	0.919186	5.456180

num_of_layer =18
예측 코드 - 궁내교

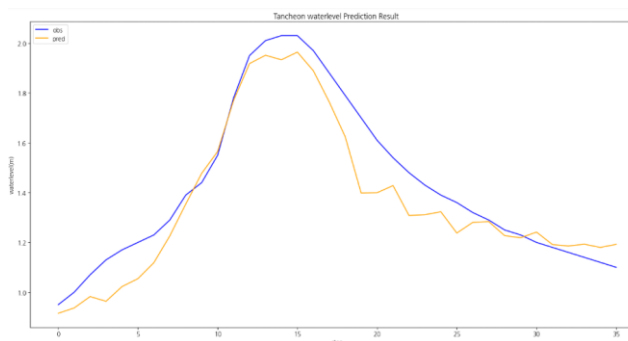
	궁내교
증설 전	testdata_220712 r2 : -0.2437545822983267 mse : 0.04935018567626799 mae : 0.18286159541871813 rmse : 0.22214901682489616
	testdata_230711 r2 : 0.8500162646896093 mse : 0.04197159430678016 mae : 0.14689570089181267 rmse : 0.2048697008021932
증설 후	testdata_250716 r2 : 0.8895288897883276 mse : 0.011082707267791009 mae : 0.08337979230615827 rmse : 0.1052744378649965
	testdata_250814 r2 : 0.7838021043271743 mse : 0.015242402057216858 mae : 0.0962281313869688 rmse : 0.1234601233484596



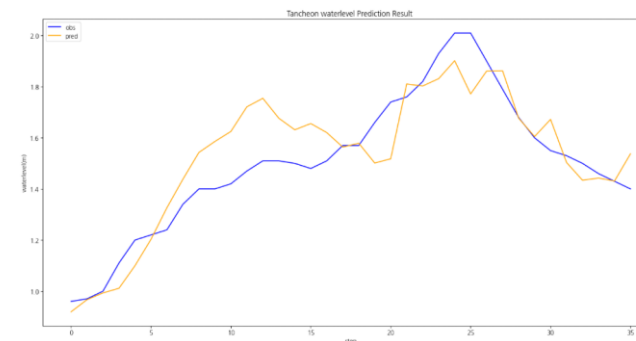
testdata_220712



testdata_230711



testdata_250716



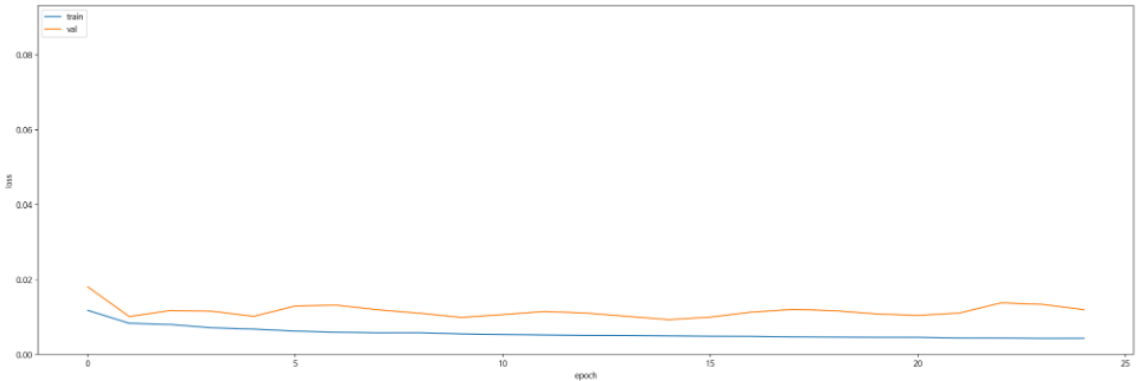
testdata_250814

**1-1. num_of_layer = 18
+ MAE**

num_of_layer =18
loss_function = MAE
학습 코드 - 공내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

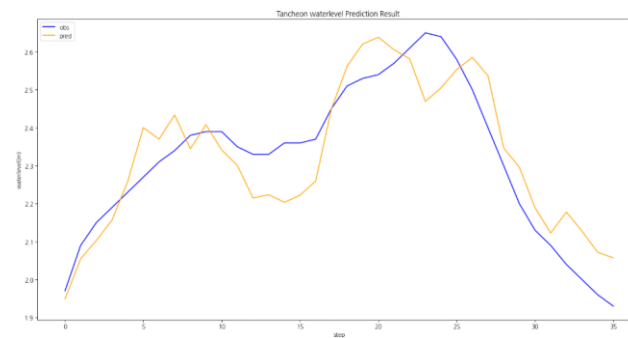
Train (23111 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0206	0.0197	0.0312	0.978130	0.980108	12.456093

Valid (6282 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0378	0.0424	0.0551	0.940861	0.952862	3.006159

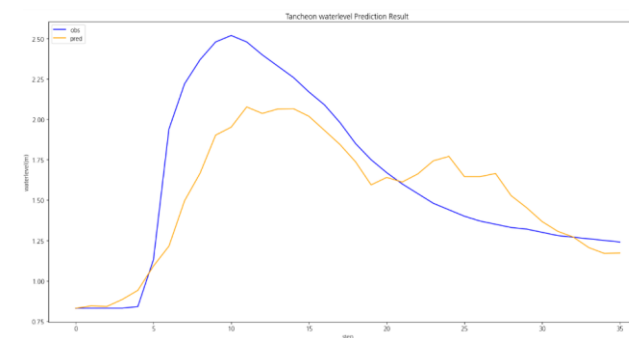
Test (5472 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0453	0.0538	0.0642	0.884159	0.915125	4.980753

num_of_layer = 18
 loss_function = MAE
 예측 코드 - 궁내교

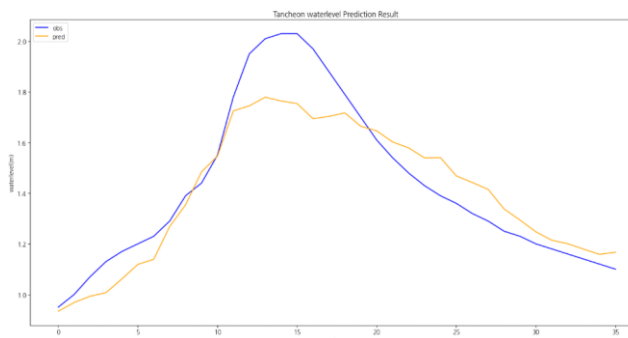
	궁내교
증설 전	testdata_220712 r2 : 0.7886506277757681 mse : 0.00838600388716136 mae : 0.07876580847634213 rmse : 0.09157512701143995
	testdata_230711 r2 : 0.6794901382392365 mse : 0.0896917913219421 mae : 0.2132765921619203 rmse : 0.29948587833475904
증설 후	testdata_250716 r2 : 0.8537863174988561 mse : 0.014668481547809194 mae : 0.09608093248473272 rmse : 0.12111350687602598
	testdata_250814 r2 : 0.7326166026610431 mse : 0.01885108656114091 mae : 0.09745151480038959 rmse : 0.13729925914272412



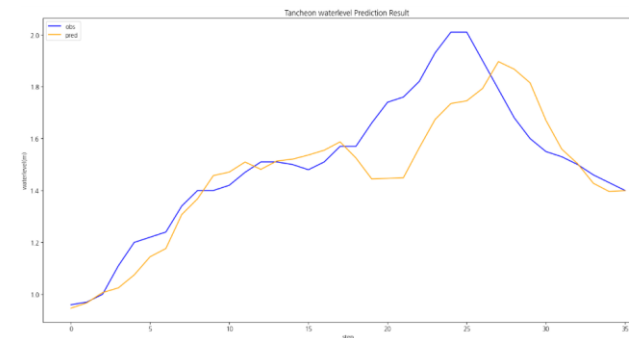
testdata_220712



testdata_230711



testdata_250716



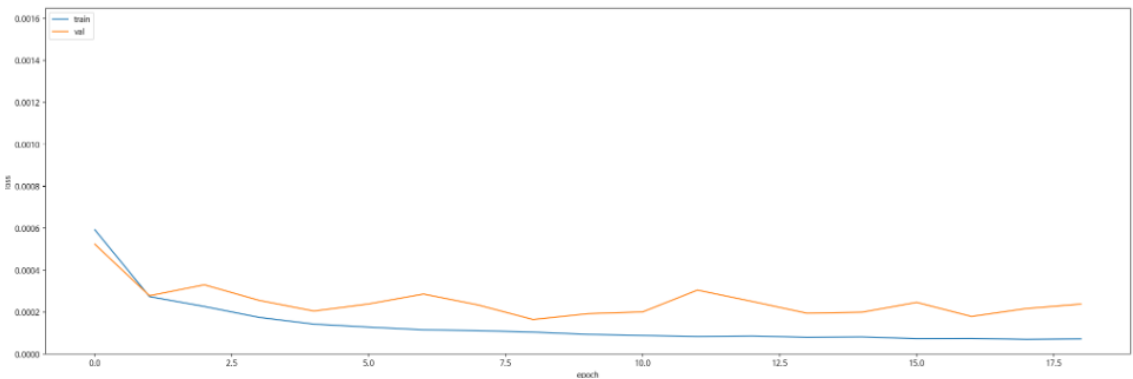
testdata_250814

2. num_of_layer = 36

num_of_layer =36
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

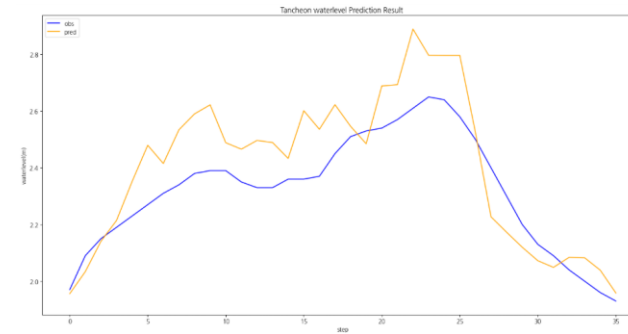
Train (23111 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0269	0.0256	0.0388	0.966277	0.970299	1.513945

Valid (6822 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0378	0.0409	0.0521	0.947172	0.956457	1.215834

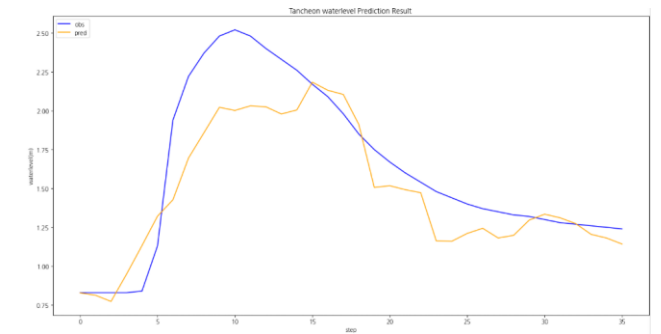
Test (5472 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0407	0.0468	0.0569	0.908998	0.930175	2.279257

num_of_layer =36
예측 코드 - 궁내교

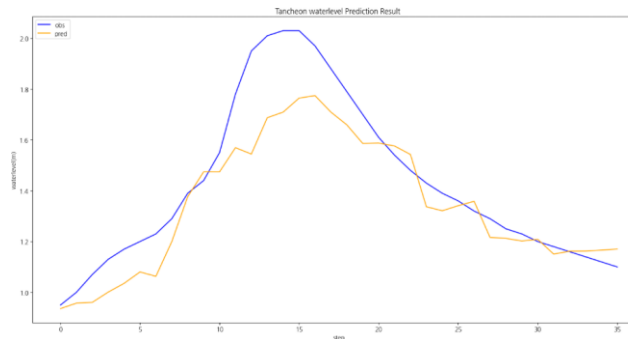
	궁내교
증설 전	testdata_220712 r2 : 0.5154125887431622 mse : 0.01922765074578906 mae : 0.11833648959795633 rmse : 0.13866380474294315
	testdata_230711 r2 : 0.7651658869889291 mse : 0.06571620649596127 mae : 0.19366505410936144 rmse : 0.2563517241915125
증설 후	testdata_250716 r2 : 0.7987440252756075 mse : 0.020190446619850436 mae : 0.10343127078480188 rmse : 0.14209309138677514
	testdata_250814 r2 : 0.9320650485021252 mse : 0.004789555611749123 mae : 0.05961067683166928 rmse : 0.06920661537562087



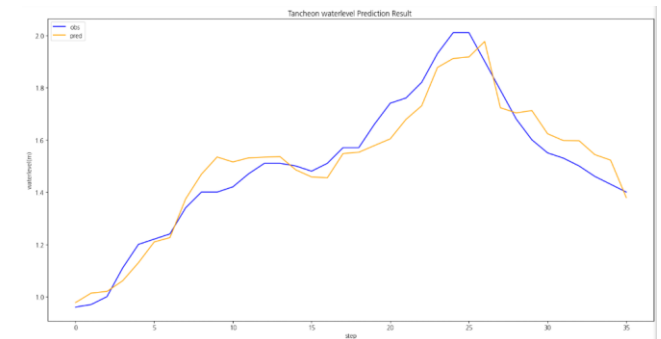
testdata_220712



testdata_230711



testdata_250716



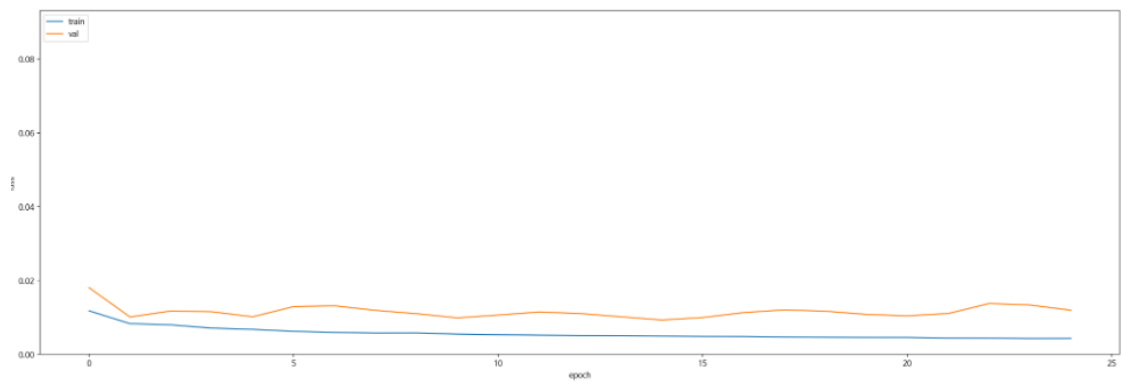
testdata_250814

Loss_function = MAE
Learning_rate = 0.0005
num_of_layer = 18

num_of_layer = 18
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (23111 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0206	0.0197	0.0312	0.978130	0.980108	12.456093

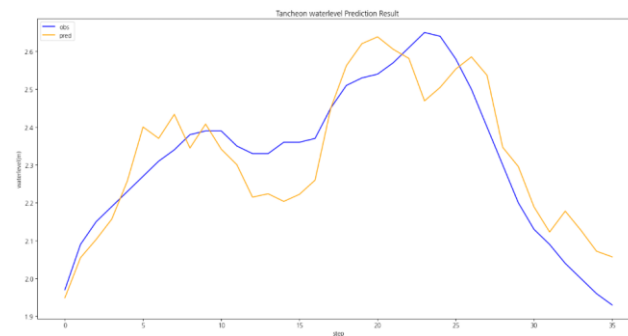
Valid (6828 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0378	0.0424	0.0551	0.940861	0.952862	3.006159

Test (5472 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0453	0.0538	0.0642	0.884159	0.915125	4.980753

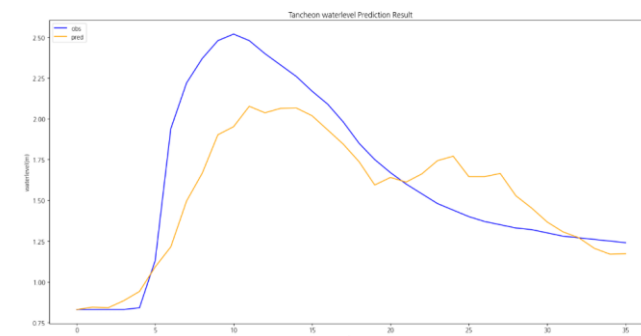
num_of_layer = 18

예측 코드 - 궁내교

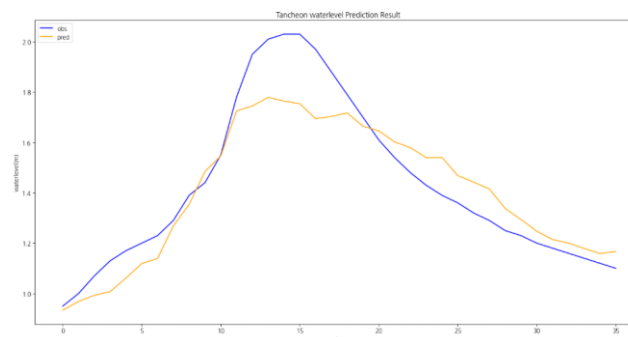
	궁내교
증설 전	testdata_220712 r2 : 0.7886506277757681 mse : 0.00838600388716136 mae : 0.07876580847634213 rmse : 0.09157512701143995
	testdata_230711 r2 : 0.6794901382392365 mse : 0.0896917913219421 mae : 0.2132765921619203 rmse : 0.29948587833475904
증설 후	testdata_250716 r2 : 0.8537863174988561 mse : 0.014668481547809194 mae : 0.09608093248473272 rmse : 0.12111350687602598
	testdata_250814 r2 : 0.7326166026610431 mse : 0.01885108656114091 mae : 0.09745151480038959 rmse : 0.13729925914272412



testdata_220712



testdata_230711



testdata_250716



testdata_250814

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 12 (각 LSTM 3개 처리)

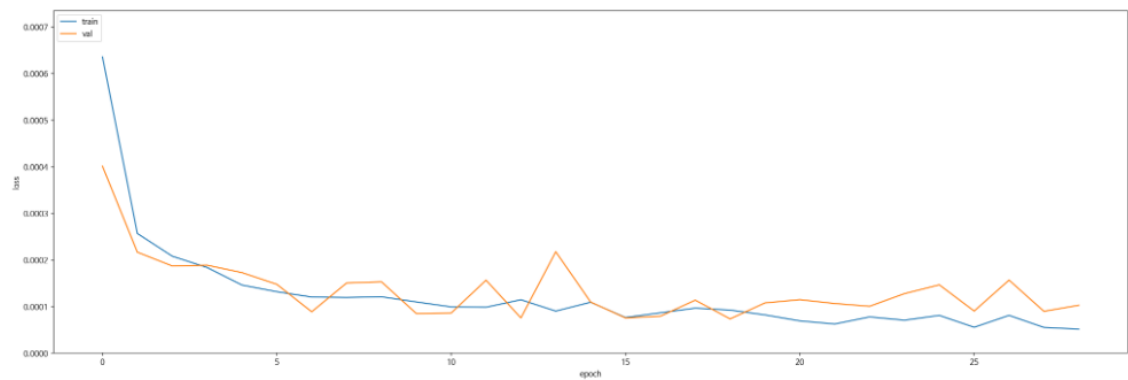
대곡교

0. num_of_layer = 12

num_of_layer =12
학습 코드 - 대곡교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

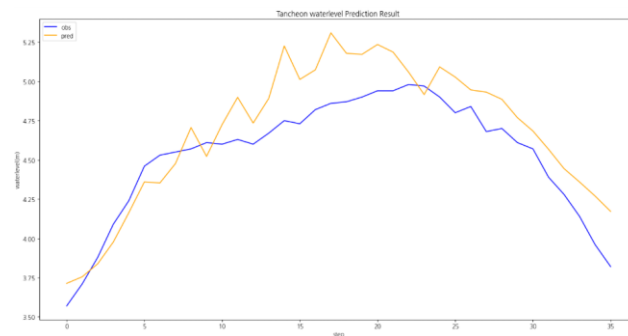
Train (23111 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0348	0.0167	0.0545	0.977872	0.981024	1.489537

Valid (6282 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0314	0.0141	0.0533	0.970796	0.973404	0.382375

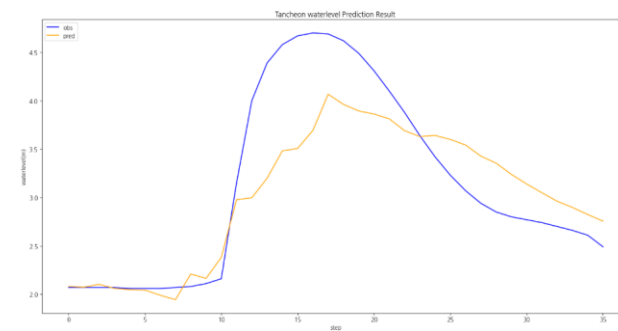
Test (5472 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0297	0.0136	0.0621	0.945628	0.946119	5.689971

num_of_layer =12
예측 코드 - 대곡교

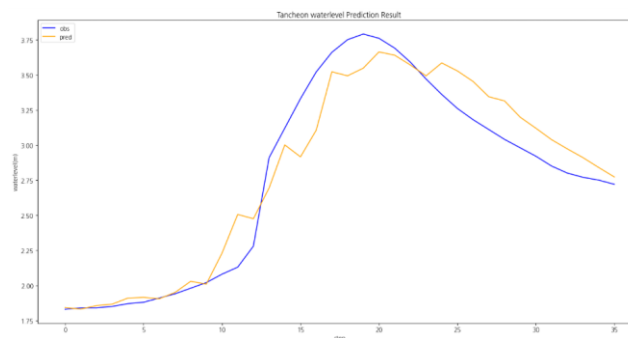
	대곡교
증설 전	testdata_220712 r2 : 0.6571355333013245 mse : 0.04801457060069652 mae : 0.19200435611936775 rmse : 0.21912227317344196
	testdata_230711 r2 : 0.7198263671371292 mse : 0.2550023667304158 mae : 0.3688772605525123 rmse : 0.5049775903249726
증설 후	testdata_250716 r2 : 0.9231141326275117 mse : 0.03550482569334603 mae : 0.14576794703801474 rmse : 0.18842724243947856
	testdata_250814 r2 : 0.8274971285050222 mse : 0.059875573660618985 mae : 0.213871206442515 rmse : 0.24469485826355034



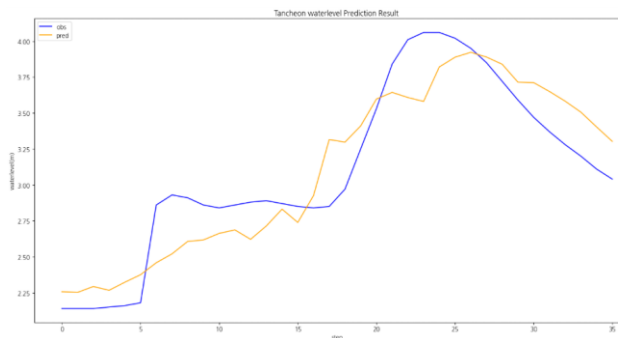
testdata_220712



testdata_230711



testdata_250716



testdata_250814